DATASCI 406 FA 26, Quiz 2.4

Name	Uniqname	UMID

Each question is worth 1 point. Circle one answer for each.

Question 1 (1 point)

$X \sim \text{Exp}(1)$, and given $X = x$, $Y \sim \text{Poisson}(x)$. Which code draws `n` pairs (X, Y) from this joint distribution?

1. `x <- rexp(n); y <- rpois(n, x)`
2. `x <- rexp(n); y <- rpois(n, rexp(n))`
3. `x <- rexp(n); y <- rpois(n, mean(x))`
4. `y <- rpois(n, 1); x <- rexp(n, y)`

Question 2 (1 point)

$\mathbf{Z} \sim N_2(\mathbf{0}, \mathbf{I}_2)$. For which matrix $\mathbf{C}$ is $\mathbf{CZ} \sim N_2\left(\mathbf{0}, \begin{pmatrix} 4 & 2 \\ 2 & 2 \end{pmatrix}\right)$? Hint: recall for matrix multiplication, $C = AB$ has $C_{ij} = \sum_k^n A_{ik} B_{kj}$.

1. $\begin{pmatrix} 4 & 0 \\ 2 & 2 \end{pmatrix}$
2. $\begin{pmatrix} 2 & 0 \\ 0 & \sqrt{2} \end{pmatrix}$
3. $\begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$
4. $\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$

Question 3 (1 point)

Why is the multivariate normal a useful model for several correlated quantities, such as test statistics computed on the same subjects?

1. It can represent any form of dependence between the quantities.
2. Its dependence is set entirely by Σ, and the CLT makes many statistics approximately normal.
3. Its components are independent, so each one can be drawn separately.
4. Probabilities such as $P(\max_i |Z_i| > c)$ have closed forms, so no simulation is needed.

Question 4 (1 point)

You want to simulate a stock's log price at the close of each trading day for the next year. Which is the best tool?

1. A Poisson process, with each arrival a trade.
2. Independent normal draws, one for each day.
3. A random walk with normal steps (Brownian motion on a daily grid).
4. A spatial Poisson process, with days as the region.